

Conducting Forensic Investigations on Linux Systems (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 06

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Time on Task:

4 hours, 42 minutes

Progress:

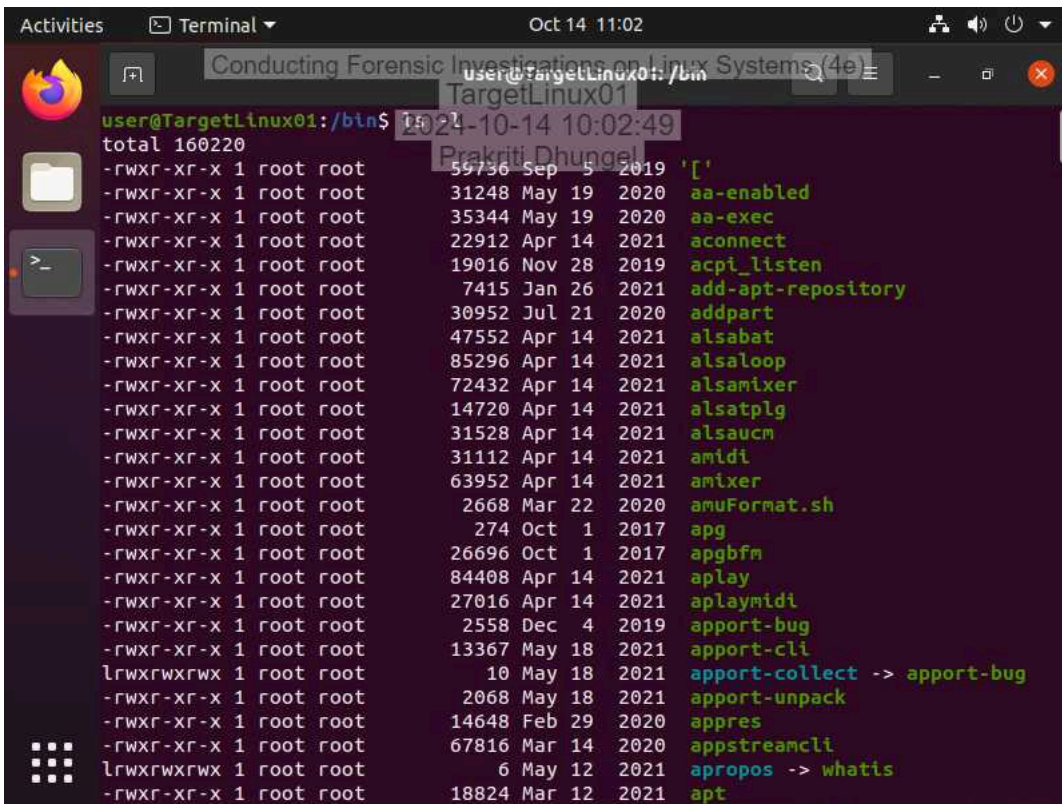
100%

Report Generated: Monday, October 14, 2024 at 1:07 PM

Section 1: Hands-On Demonstration

Part 1: Explore a Live Linux System

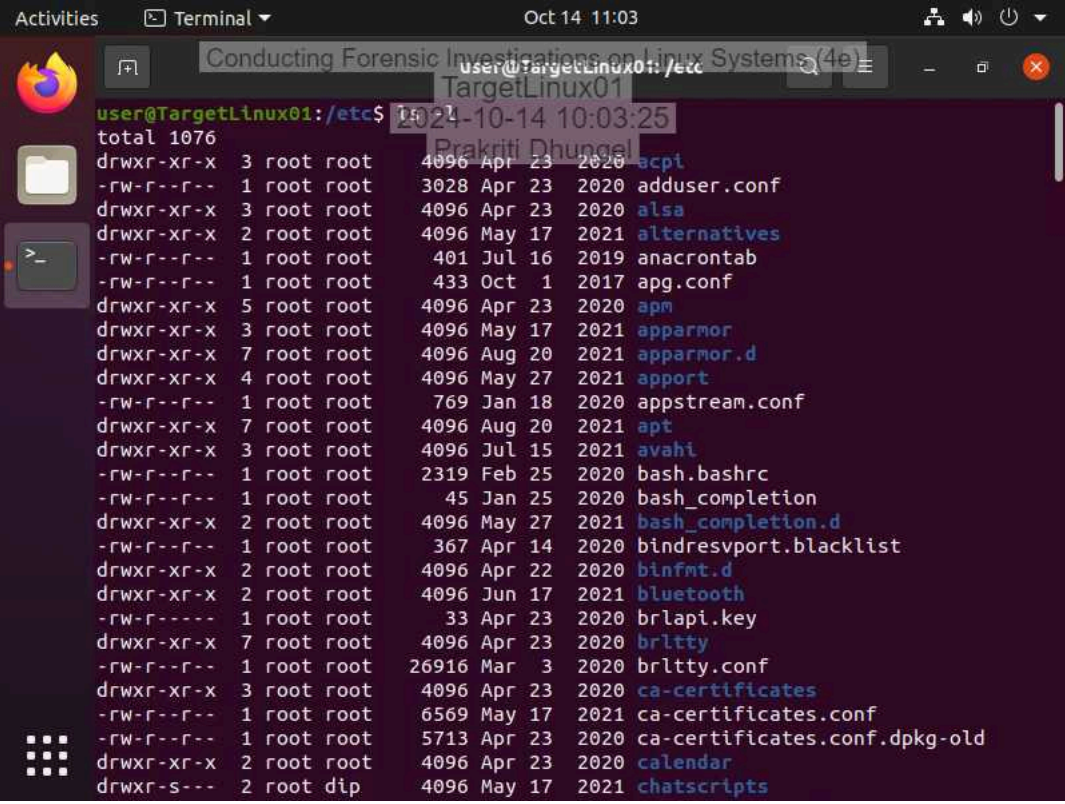
17. Make a screen capture showing the contents of the `/bin` directory.



The screenshot shows a terminal window titled "Terminal" with the date and time "Oct 14 11:02". The terminal prompt is `user@TargetLinux01: /bin$`. The command `ls -l` has been executed, displaying a long list of files in the `/bin` directory. The output shows the permissions, owner, group, size, date, and filename for each file. The files listed include `aa-enabled`, `aa-exec`, `aconect`, `acpi_listen`, `add-apt-repository`, `addpart`, `alsabat`, `alsaloop`, `alsamixer`, `alsatplg`, `alsaucm`, `amidi`, `amixer`, `amuFormat.sh`, `apg`, `apgbfm`, `aplay`, `aplaymidi`, `apport-bug`, `apport-cli`, `apport-collect`, `apport-unpack`, `appres`, `appstreamcli`, `apropos`, and `apt`. The terminal window also shows the system's desktop environment with icons for Activities, Terminal, and a file manager.

```
user@TargetLinux01: /bin$ ls -l
total 160220
-rwxr-xr-x 1 root root 59736 Sep  5 2019 '['
-rwxr-xr-x 1 root root 31248 May 19 2020 aa-enabled
-rwxr-xr-x 1 root root 35344 May 19 2020 aa-exec
-rwxr-xr-x 1 root root 22912 Apr 14 2021 aconect
-rwxr-xr-x 1 root root 19016 Nov 28 2019 acpi_listen
-rwxr-xr-x 1 root root 7415 Jan 26 2021 add-apt-repository
-rwxr-xr-x 1 root root 30952 Jul 21 2020 addpart
-rwxr-xr-x 1 root root 47552 Apr 14 2021 alsabat
-rwxr-xr-x 1 root root 85296 Apr 14 2021 alsaloop
-rwxr-xr-x 1 root root 72432 Apr 14 2021 alsamixer
-rwxr-xr-x 1 root root 14720 Apr 14 2021 alsatplg
-rwxr-xr-x 1 root root 31528 Apr 14 2021 alsaucm
-rwxr-xr-x 1 root root 31112 Apr 14 2021 amidi
-rwxr-xr-x 1 root root 63952 Apr 14 2021 amixer
-rwxr-xr-x 1 root root 2668 Mar 22 2020 amuFormat.sh
-rwxr-xr-x 1 root root 274 Oct  1 2017 apg
-rwxr-xr-x 1 root root 26696 Oct  1 2017 apgbfm
-rwxr-xr-x 1 root root 84408 Apr 14 2021 aplay
-rwxr-xr-x 1 root root 27016 Apr 14 2021 aplaymidi
-rwxr-xr-x 1 root root 2558 Dec  4 2019 apport-bug
-rwxr-xr-x 1 root root 13367 May 18 2021 apport-cli
lrwxrwxrwx 1 root root 10 May 18 2021 apport-collect -> apport-bug
-rwxr-xr-x 1 root root 2068 May 18 2021 apport-unpack
-rwxr-xr-x 1 root root 14648 Feb 29 2020 appres
-rwxr-xr-x 1 root root 67816 Mar 14 2020 appstreamcli
lrwxrwxrwx 1 root root 6 May 12 2021 apropos -> whatis
-rwxr-xr-x 1 root root 18824 Mar 12 2021 apt
```

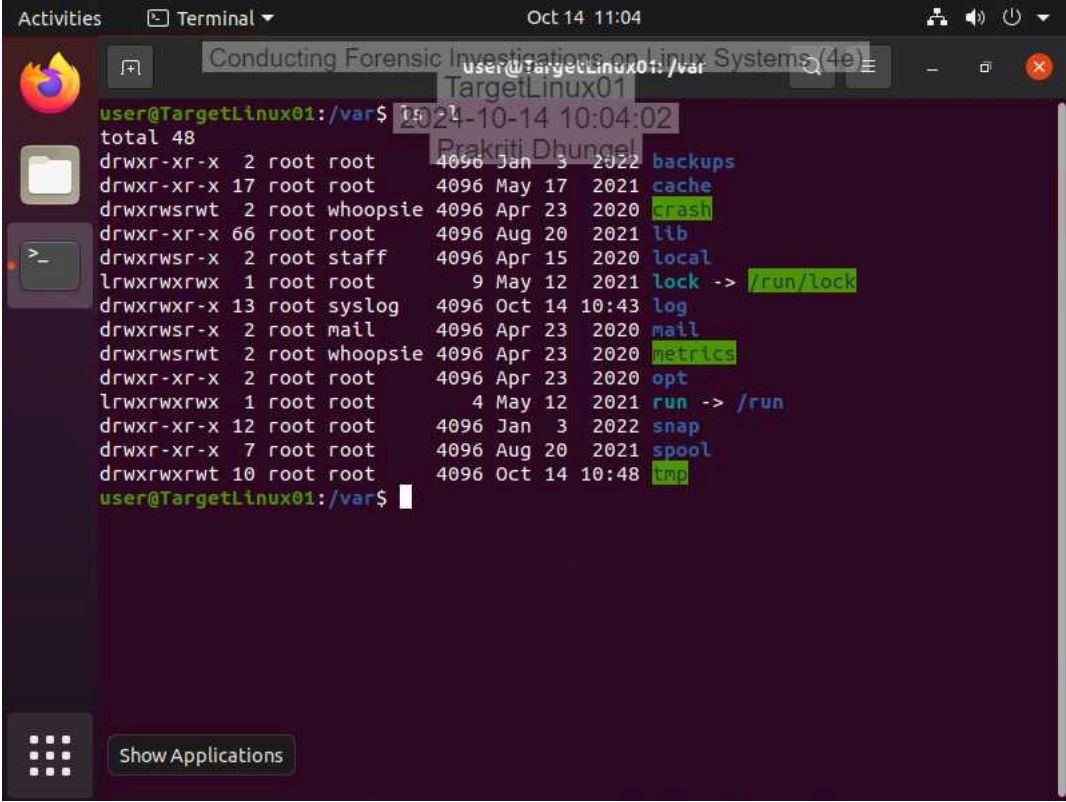
20. Make a screen capture showing the contents of the /etc directory.



A screenshot of a Linux terminal window titled "Terminal" showing the output of the `ls -l` command in the `/etc` directory. The window's title bar includes "Activities", "Terminal", and the date/time "Oct 14 11:03". The terminal prompt is `user@TargetLinux01:/etc$`. The output lists files with their permissions, owner, group, size, and modification date. The files listed are: `acpi`, `adduser.conf`, `alsa`, `alternatives`, `anacrontab`, `apg.conf`, `apm`, `apparmor`, `apparmor.d`, `appport`, `appstream.conf`, `apt`, `avahi`, `bash.bashrc`, `bash_completion`, `bash_completion.d`, `bindresvport.blacklist`, `binfmt.d`, `bluetooth`, `brlapi.key`, `brltty`, `brltty.conf`, `ca-certificates`, `ca-certificates.conf`, `ca-certificates.conf.dpkg-old`, `calendar`, and `chatscripts`. The window also shows a sidebar with icons for Activities, Files, and Terminal, and a top bar with the title "Conducting Forensic Investigations on Linux Systems (4e)" and a search bar.

```
total 1076
drwxr-xr-x 3 root root 4096 Apr 23 2020 acpi
-rw-r--r-- 1 root root 3028 Apr 23 2020 adduser.conf
drwxr-xr-x 3 root root 4096 Apr 23 2020 alsa
drwxr-xr-x 2 root root 4096 May 17 2021 alternatives
-rw-r--r-- 1 root root 401 Jul 16 2019 anacrontab
-rw-r--r-- 1 root root 433 Oct 1 2017 apg.conf
drwxr-xr-x 5 root root 4096 Apr 23 2020 apm
drwxr-xr-x 3 root root 4096 May 17 2021 apparmor
drwxr-xr-x 7 root root 4096 Aug 20 2021 apparmor.d
drwxr-xr-x 4 root root 4096 May 27 2021 appport
-rw-r--r-- 1 root root 769 Jan 18 2020 appstream.conf
drwxr-xr-x 7 root root 4096 Aug 20 2021 apt
drwxr-xr-x 3 root root 4096 Jul 15 2021 avahi
-rw-r--r-- 1 root root 2319 Feb 25 2020 bash.bashrc
-rw-r--r-- 1 root root 45 Jan 25 2020 bash_completion
drwxr-xr-x 2 root root 4096 May 27 2021 bash_completion.d
-rw-r--r-- 1 root root 367 Apr 14 2020 bindresvport.blacklist
drwxr-xr-x 2 root root 4096 Apr 22 2020 binfmt.d
drwxr-xr-x 2 root root 4096 Jun 17 2021 bluetooth
-rw-r--r-- 1 root root 33 Apr 23 2020 brlapi.key
drwxr-xr-x 7 root root 4096 Apr 23 2020 brltty
-rw-r--r-- 1 root root 26916 Mar 3 2020 brltty.conf
drwxr-xr-x 3 root root 4096 Apr 23 2020 ca-certificates
-rw-r--r-- 1 root root 6569 May 17 2021 ca-certificates.conf
-rw-r--r-- 1 root root 5713 Apr 23 2020 ca-certificates.conf.dpkg-old
drwxr-xr-x 2 root root 4096 Apr 23 2020 calendar
drwxr-s-- 2 root dip 4096 May 17 2021 chatscripts
```

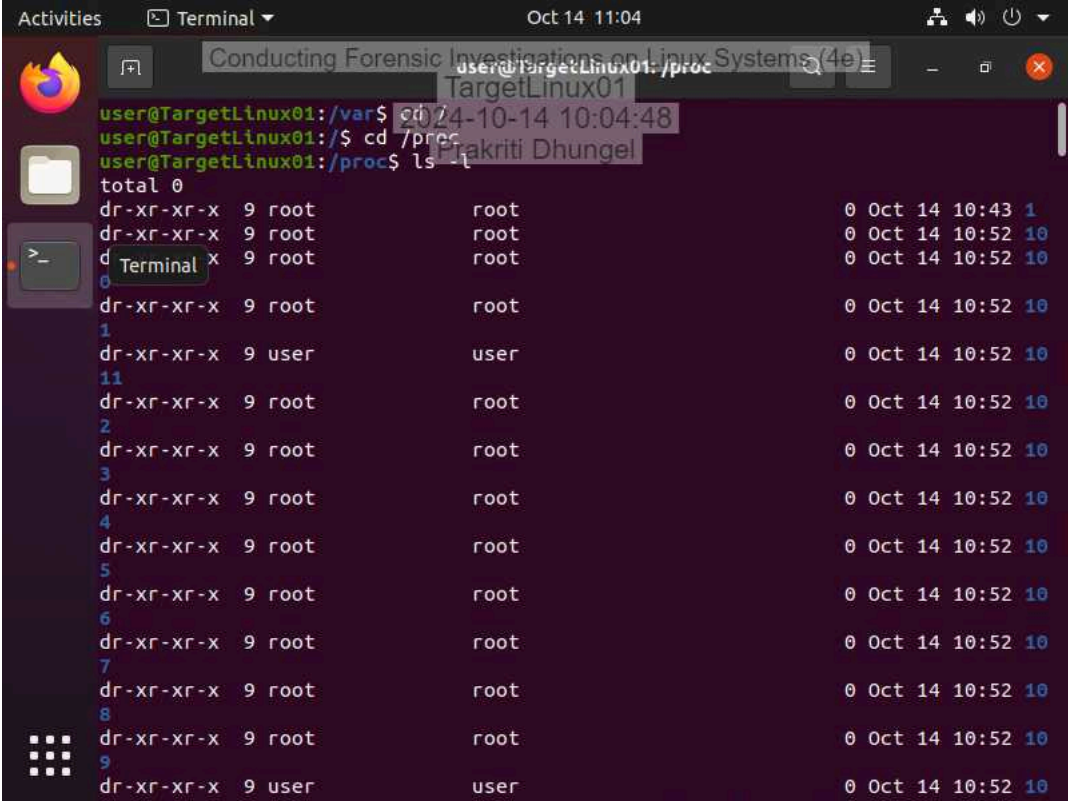
21. Make a screen capture showing the contents of the /var directory.



A terminal window titled "Terminal" showing the command `ls -l /var` and its output. The output lists the contents of the /var directory, including subdirectories like backups, cache, crash, lib, local, lock, log, mail, metrics, opt, run, snap, spool, and tmp. The permissions, owner, and group for each directory are also displayed. The terminal window is part of a desktop environment with a sidebar on the left showing icons for Activities, Files, and a terminal icon. The top of the window shows the date and time as "Oct 14 11:04".

```
user@TargetLinux01: /var$ ls -l /var
total 48
drwxr-xr-x  2 root root    4096 Jan  3  2022 backups
drwxr-xr-x 17 root root    4096 May 17  2021 cache
drwxrwsrwt  2 root whoopsie 4096 Apr 23  2020 crash
drwxr-xr-x 66 root root    4096 Aug 20  2021 lib
drwxrwsr-x  2 root staff   4096 Apr 15  2020 local
lrwxrwxrwx  1 root root      9 May 12  2021 lock -> /run/lock
drwxrwsr-x 13 root syslog  4096 Oct 14 10:43 log
drwxrwsr-x  2 root mail    4096 Apr 23  2020 mail
drwxrwsrwt  2 root whoopsie 4096 Apr 23  2020 metrics
drwxr-xr-x  2 root root    4096 Apr 23  2020 opt
lrwxrwxrwx  1 root root      4 May 12  2021 run -> /run
drwxr-xr-x 12 root root    4096 Jan  3  2022 snap
drwxr-xr-x  7 root root    4096 Aug 20  2021 spool
drwxrwsrwt 10 root root    4096 Oct 14 10:48 tmp
user@TargetLinux01: /var$
```

22. Make a screen capture showing the contents of the /proc directory.

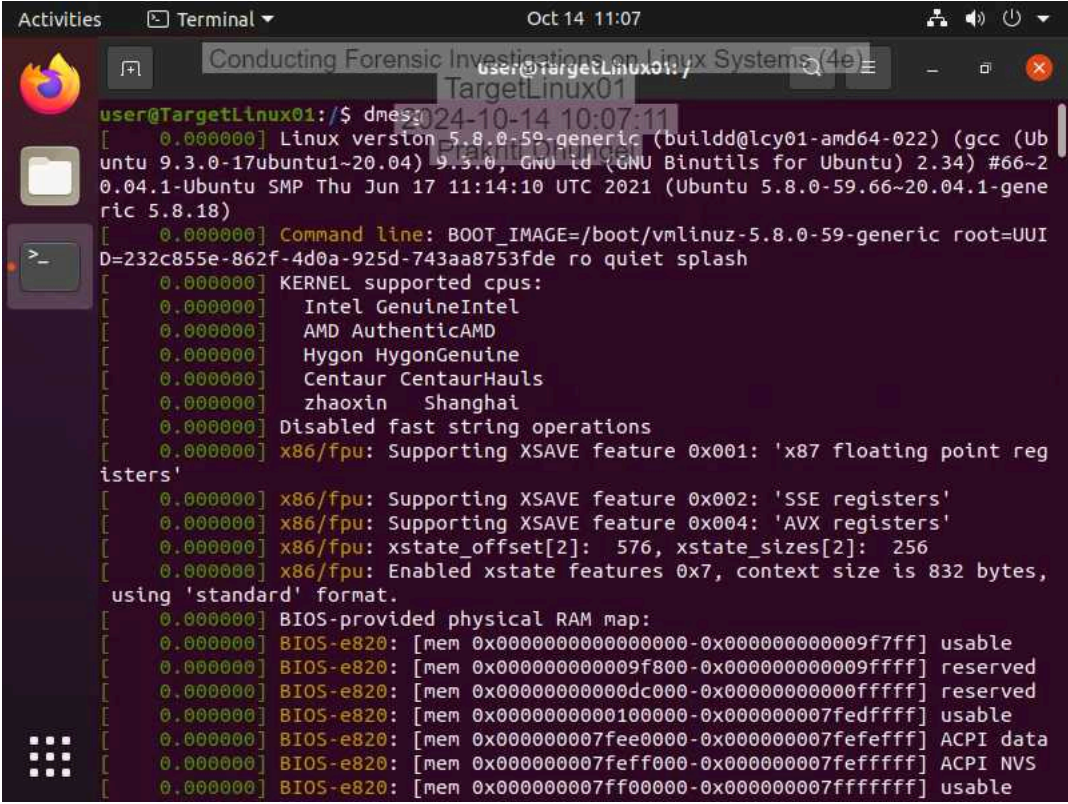


The screenshot shows a terminal window titled "Terminal" with the date and time "Oct 14 11:04". The terminal output shows the user navigating to the /proc directory and listing its contents. The output is as follows:

```
user@TargetLinux01:/var$ cd /
user@TargetLinux01:/ $ cd /proc
user@TargetLinux01:/proc$ ls -lt
total 0
dr-xr-xr-x  9 root          root          0 Oct 14 10:43 1
dr-xr-xr-x  9 root          root          0 Oct 14 10:52 10
d Terminal x  9 root          root          0 Oct 14 10:52 10
0
dr-xr-xr-x  9 root          root          0 Oct 14 10:52 10
1
dr-xr-xr-x  9 user          user          0 Oct 14 10:52 10
11
dr-xr-xr-x  9 root          root          0 Oct 14 10:52 10
2
dr-xr-xr-x  9 root          root          0 Oct 14 10:52 10
3
dr-xr-xr-x  9 root          root          0 Oct 14 10:52 10
4
dr-xr-xr-x  9 root          root          0 Oct 14 10:52 10
5
dr-xr-xr-x  9 root          root          0 Oct 14 10:52 10
6
dr-xr-xr-x  9 root          root          0 Oct 14 10:52 10
7
dr-xr-xr-x  9 root          root          0 Oct 14 10:52 10
8
dr-xr-xr-x  9 root          root          0 Oct 14 10:52 10
9
dr-xr-xr-x  9 user          user          0 Oct 14 10:52 10
```

Part 2: Use Linux Shell Commands for Forensic Investigations

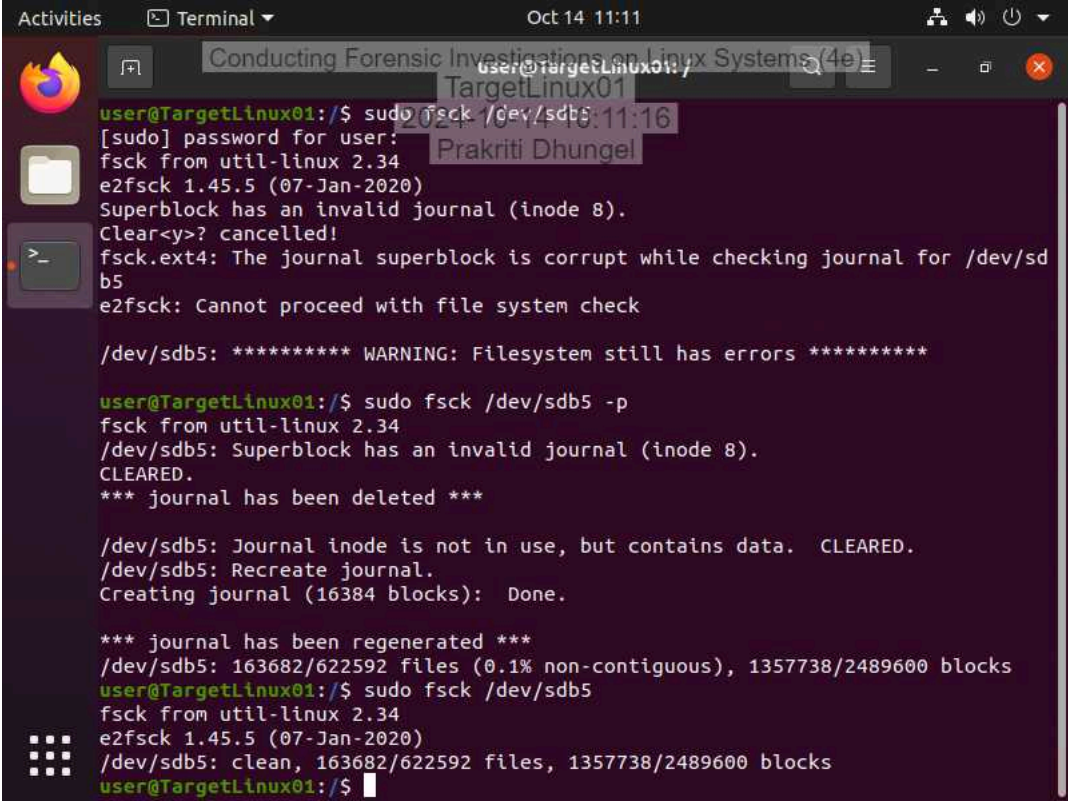
2. Make a screen capture showing the results of the `dmesg` command.



The screenshot shows a terminal window titled "Terminal" with the date and time "Oct 14 11:07". The user is logged in as "user" on a system named "TargetLinux01". The terminal displays the output of the `dmesg` command, which shows system boot logs. The output includes the Linux version (5.8.0-59-generic), the command line, kernel supported CPUs (Intel GenuineIntel, AMD AuthenticAMD, Hygon HygonGenuine, Centaur CentaurHauls, zhaoxin Shanghai), disabled fast string operations, x86/fpu supporting XSAVE features (SSE registers, AVX registers), and the BIOS-provided physical RAM map.

```
user@TargetLinux01:/$ dmesg
[ 0.000000] Linux version 5.8.0-59-generic (buildd@lcy01-amd64-022) (gcc (Ubuntu 9.3.0-17ubuntu1~20.04) 9.3.0, GNU ld (GNU Binutils for Ubuntu) 2.34) #66~20.04.1-Ubuntu SMP Thu Jun 17 11:14:10 UTC 2021 (Ubuntu 5.8.0-59.66~20.04.1-generic 5.8.18)
[ 0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-5.8.0-59-generic root=UUID=D=232c855e-862f-4d0a-925d-743aa8753fde ro quiet splash
[ 0.000000] KERNEL supported cpus:
[ 0.000000]   Intel GenuineIntel
[ 0.000000]   AMD AuthenticAMD
[ 0.000000]   Hygon HygonGenuine
[ 0.000000]   Centaur CentaurHauls
[ 0.000000]   zhaoxin   Shanghai
[ 0.000000] Disabled fast string operations
[ 0.000000] x86/fpu: Supporting XSAVE feature 0x001: 'x87 floating point registers'
[ 0.000000] x86/fpu: Supporting XSAVE feature 0x002: 'SSE registers'
[ 0.000000] x86/fpu: Supporting XSAVE feature 0x004: 'AVX registers'
[ 0.000000] x86/fpu: xstate_offset[2]: 576, xstate_sizes[2]: 256
[ 0.000000] x86/fpu: Enabled xstate features 0x7, context size is 832 bytes, using 'standard' format.
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x00000000000009f7ff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000009f800-0x00000000000009ffff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000000dc000-0x0000000000000fffff] reserved
[ 0.000000] BIOS-e820: [mem 0x000000000000100000-0x00000000000007fedffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000000007fee0000-0x00000000000007fefefff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000007feff000-0x00000000000007fefffff] ACPI NVS
[ 0.000000] BIOS-e820: [mem 0x0000000000007ff00000-0x00000000000007ffffffffff] usable
```

7. Make a screen capture showing the results of the fsck command.



A terminal window titled "Terminal" with a date and time of "Oct 14 11:11". The window shows the execution of the fsck command on /dev/sdb5. The output indicates a corrupted journal superblock and a warning that the filesystem still has errors. The user then runs fsck with the -p flag, which successfully regenerates the journal and reports the filesystem as clean.

```
user@TargetLinux01:/$ sudo fsck /dev/sdb5
[sudo] password for user:
fsck from util-linux 2.34
e2fsck 1.45.5 (07-Jan-2020)
Superblock has an invalid journal (inode 8).
Clear<y>? cancelled!
fsck.ext4: The journal superblock is corrupt while checking journal for /dev/sd
b5
e2fsck: Cannot proceed with file system check

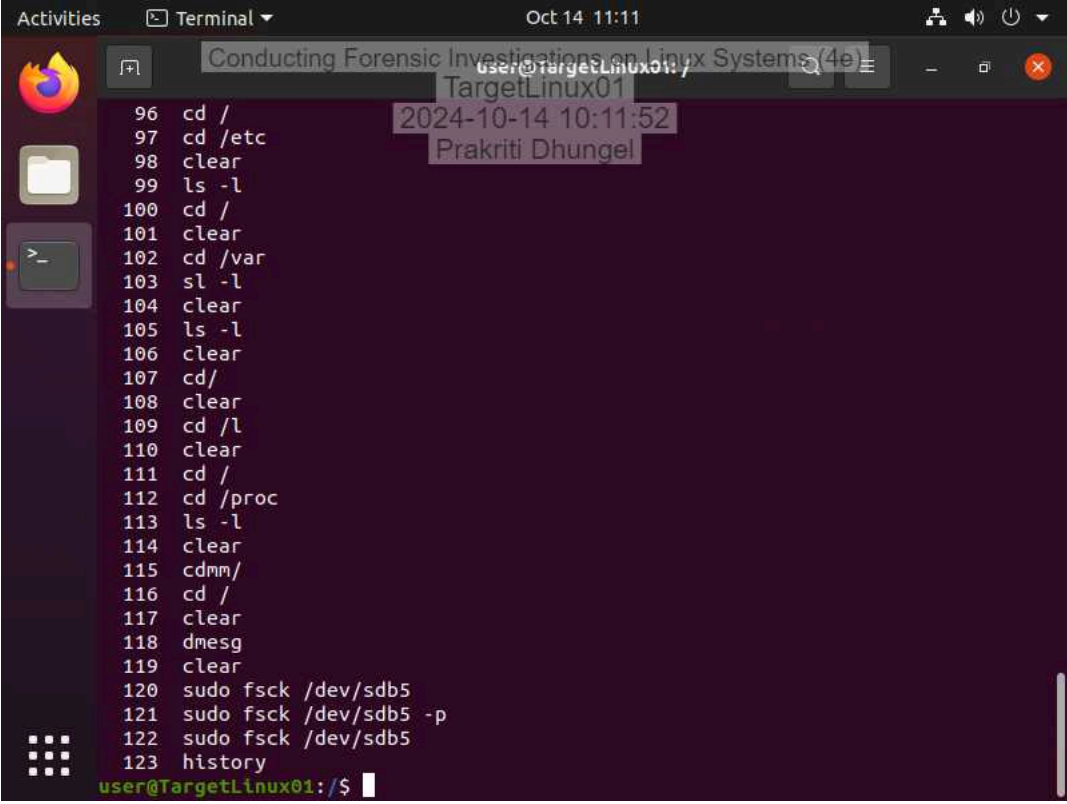
/dev/sdb5: ***** WARNING: Filesystem still has errors *****

user@TargetLinux01:/$ sudo fsck /dev/sdb5 -p
fsck from util-linux 2.34
/dev/sdb5: Superblock has an invalid journal (inode 8).
CLEARED.
*** journal has been deleted ***

/dev/sdb5: Journal inode is not in use, but contains data.  CLEARED.
/dev/sdb5: Recreate journal.
Creating journal (16384 blocks):  Done.

*** journal has been regenerated ***
/dev/sdb5: 163682/622592 files (0.1% non-contiguous), 1357738/2489600 blocks
user@TargetLinux01:/$ sudo fsck /dev/sdb5
fsck from util-linux 2.34
e2fsck 1.45.5 (07-Jan-2020)
/dev/sdb5: clean, 163682/622592 files, 1357738/2489600 blocks
user@TargetLinux01:/$
```

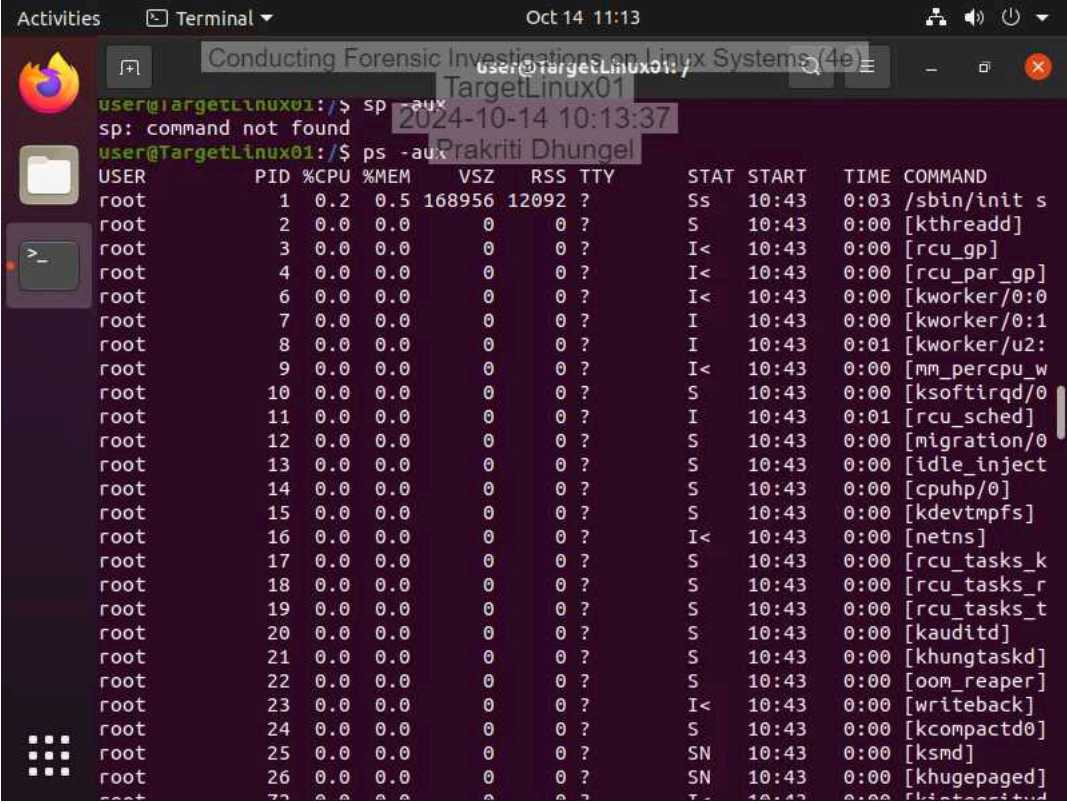
9. Make a screen capture showing the results of the history command.



```
Oct 14 11:11
user@TargetLinux01: /
2024-10-14 10:11:52
Prakriti Dhungel

96 cd /
97 cd /etc
98 clear
99 ls -l
100 cd /
101 clear
102 cd /var
103 sl -l
104 clear
105 ls -l
106 clear
107 cd/
108 clear
109 cd /l
110 clear
111 cd /
112 cd /proc
113 ls -l
114 clear
115 cdmm/
116 cd /
117 clear
118 dmesg
119 clear
120 sudo fsck /dev/sdb5
121 sudo fsck /dev/sdb5 -p
122 sudo fsck /dev/sdb5
123 history
user@TargetLinux01:/$
```

11. Make a screen capture showing the running processes.

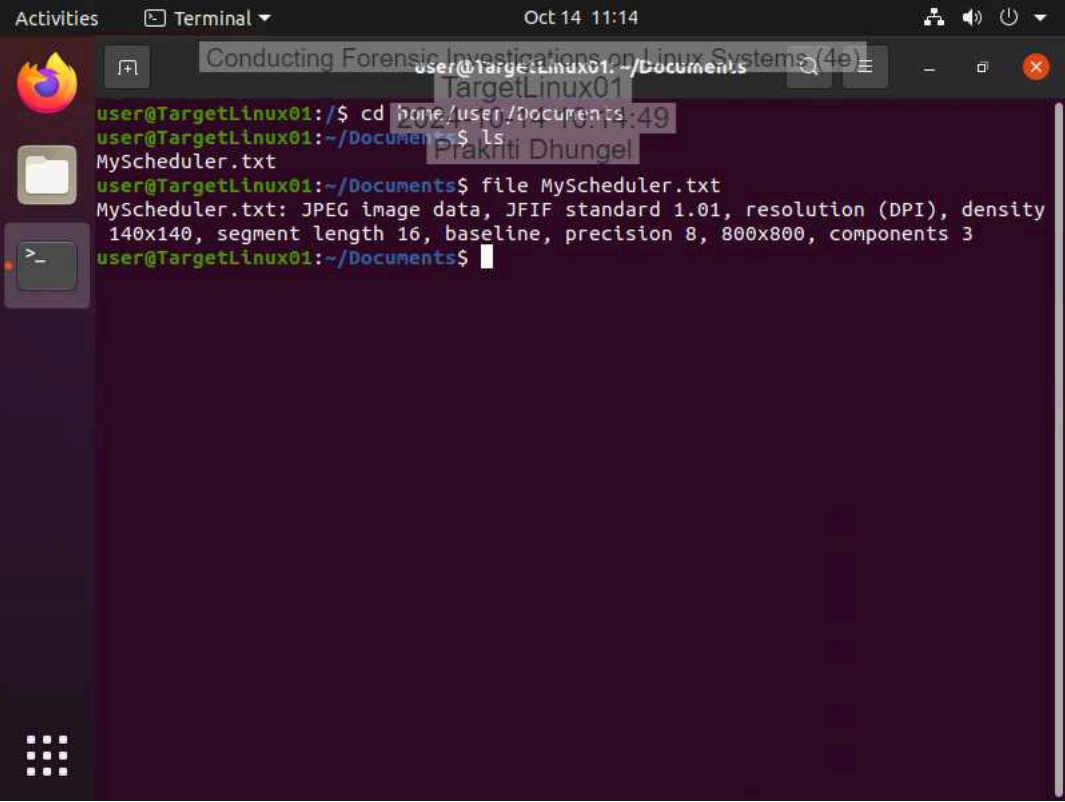


A terminal window titled "Terminal" with a date and time of "Oct 14 11:13". The terminal shows a user prompt "user@TargetLinux01:/" followed by the command "ps -aux". The output is a table of running processes. The terminal also shows a search bar at the top with the text "Conducting Forensic Investigations on Linux Systems (4e)" and a date stamp "2024-10-14 10:13:37".

```
user@TargetLinux01:/$ ps -aux
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.2	0.5	168956	12092	?	Ss	10:43	0:03	/sbin/init s
root	2	0.0	0.0	0	0	?	S	10:43	0:00	[kthreadd]
root	3	0.0	0.0	0	0	?	I<	10:43	0:00	[rcu_gp]
root	4	0.0	0.0	0	0	?	I<	10:43	0:00	[rcu_par_gp]
root	6	0.0	0.0	0	0	?	I<	10:43	0:00	[kworker/0:0
root	7	0.0	0.0	0	0	?	I	10:43	0:00	[kworker/0:1
root	8	0.0	0.0	0	0	?	I	10:43	0:01	[kworker/u2:
root	9	0.0	0.0	0	0	?	I<	10:43	0:00	[mm_percpu_w
root	10	0.0	0.0	0	0	?	S	10:43	0:00	[ksoftirqd/0
root	11	0.0	0.0	0	0	?	I	10:43	0:01	[rcu_sched]
root	12	0.0	0.0	0	0	?	S	10:43	0:00	[migration/0
root	13	0.0	0.0	0	0	?	S	10:43	0:00	[idle_inject
root	14	0.0	0.0	0	0	?	S	10:43	0:00	[cpuhp/0]
root	15	0.0	0.0	0	0	?	S	10:43	0:00	[kdevtmpfs]
root	16	0.0	0.0	0	0	?	I<	10:43	0:00	[netns]
root	17	0.0	0.0	0	0	?	S	10:43	0:00	[rcu_tasks_k
root	18	0.0	0.0	0	0	?	S	10:43	0:00	[rcu_tasks_r
root	19	0.0	0.0	0	0	?	S	10:43	0:00	[rcu_tasks_t
root	20	0.0	0.0	0	0	?	S	10:43	0:00	[kauditd]
root	21	0.0	0.0	0	0	?	S	10:43	0:00	[khungtaskd]
root	22	0.0	0.0	0	0	?	S	10:43	0:00	[oom_reaper]
root	23	0.0	0.0	0	0	?	I<	10:43	0:00	[writeback]
root	24	0.0	0.0	0	0	?	S	10:43	0:00	[kcompactd0]
root	25	0.0	0.0	0	0	?	SN	10:43	0:00	[ksmd]
root	26	0.0	0.0	0	0	?	SN	10:43	0:00	[khugepaged]
root	32	0.0	0.0	0	0	?	T<	10:43	0:00	[kintegrityd

15. Make a screen capture showing the results of the file command.

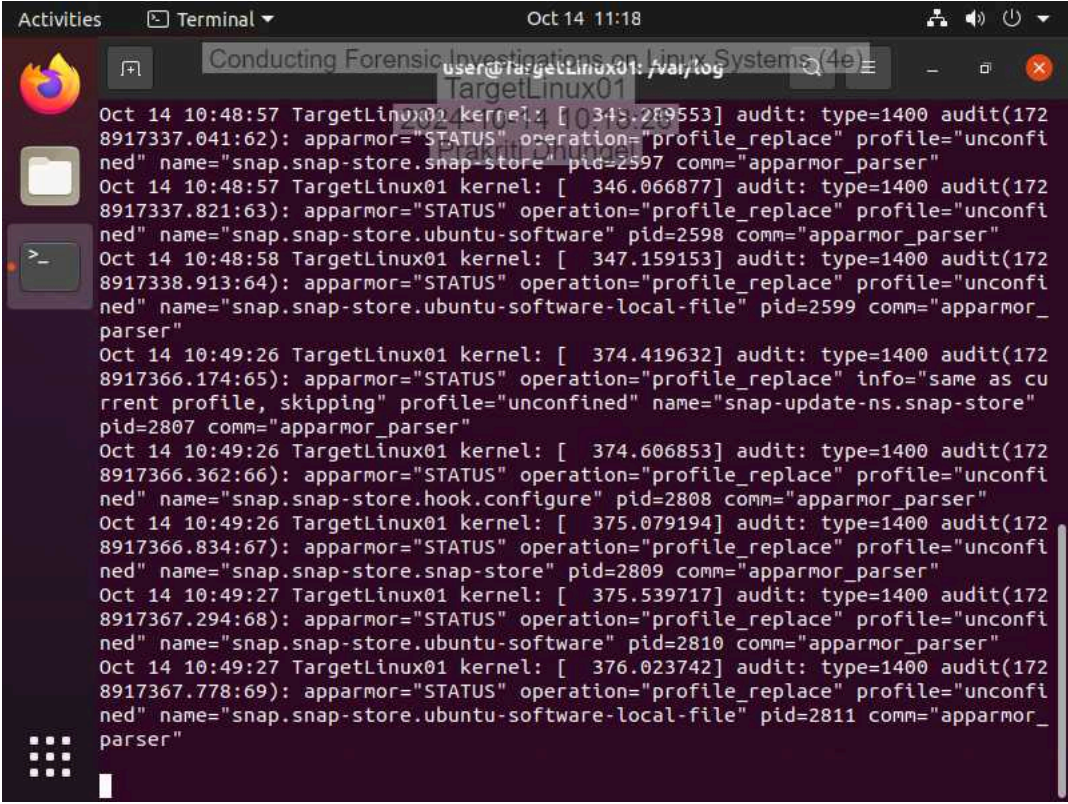


A terminal window titled "Terminal" with a subtitle "Conducting Forensic Investigations on Linux Systems (4e)" is shown. The window has a dark background and a light-colored text. The terminal output is as follows:

```
user@TargetLinux01: ~/Documents
user@TargetLinux01:~/Documents$ cd /home/user/Documents
user@TargetLinux01:~/Documents$ ls
MyScheduler.txt
user@TargetLinux01:~/Documents$ file MyScheduler.txt
MyScheduler.txt: JPEG image data, JFIF standard 1.01, resolution (DPI), density
140x140, segment length 16, baseline, precision 8, 800x800, components 3
user@TargetLinux01:~/Documents$
```

Part 3: Retrieve Logs Files on a Live Linux System

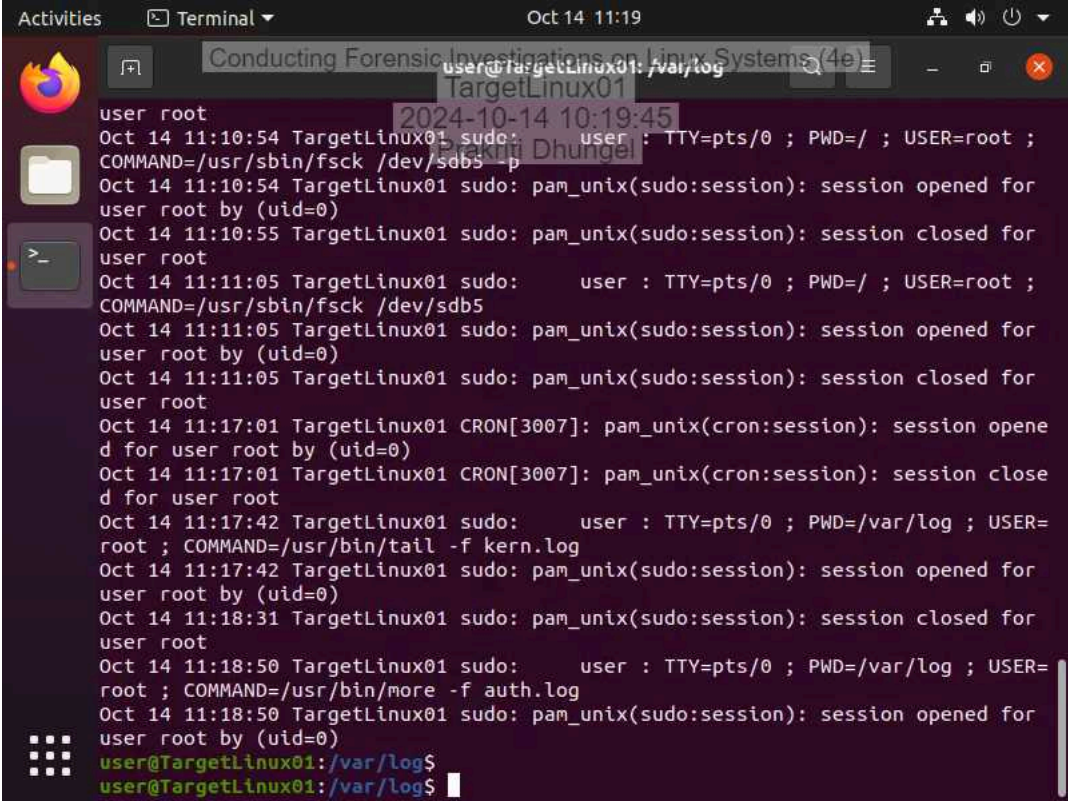
4. Make a screen capture showing the records in the kern.log file.



The screenshot shows a terminal window titled "Conducting Forensic Investigations on Linux Systems (4e)" with the prompt "user@TargetLinux01: /var/log". The terminal displays the contents of the /var/log/kern.log file, showing a series of audit records. Each record starts with a timestamp and the source "TargetLinux01 kernel:", followed by kernel parameters in brackets and an audit message. The audit messages are of type 1400 and contain details about apparmor profile replacements for various snap components.

```
Oct 14 10:48:57 TargetLinux01 kernel: [ 345.269553] audit: type=1400 audit(1728917337.041:62): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=2597 comm="apparmor_parser"
Oct 14 10:48:57 TargetLinux01 kernel: [ 346.066877] audit: type=1400 audit(1728917337.821:63): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=2598 comm="apparmor_parser"
Oct 14 10:48:58 TargetLinux01 kernel: [ 347.159153] audit: type=1400 audit(1728917338.913:64): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software-local-file" pid=2599 comm="apparmor_parser"
Oct 14 10:49:26 TargetLinux01 kernel: [ 374.419632] audit: type=1400 audit(1728917366.174:65): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="snap-update-ns.snap-store" pid=2807 comm="apparmor_parser"
Oct 14 10:49:26 TargetLinux01 kernel: [ 374.606853] audit: type=1400 audit(1728917366.362:66): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.hook.configure" pid=2808 comm="apparmor_parser"
Oct 14 10:49:26 TargetLinux01 kernel: [ 375.079194] audit: type=1400 audit(1728917366.834:67): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=2809 comm="apparmor_parser"
Oct 14 10:49:27 TargetLinux01 kernel: [ 375.539717] audit: type=1400 audit(1728917367.294:68): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=2810 comm="apparmor_parser"
Oct 14 10:49:27 TargetLinux01 kernel: [ 376.023742] audit: type=1400 audit(1728917367.778:69): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software-local-file" pid=2811 comm="apparmor_parser"
```

7. Make a screen capture showing the records in the auth.log file.

A terminal window titled 'Terminal' with a date and time of 'Oct 14 11:19'. The window shows the output of the 'cat /var/log/auth.log' command. The output displays several log entries for user root, including session opening and closing, and commands executed like 'fsck', 'tail -f kern.log', and 'more -f auth.log'. The prompt 'user@TargetLinux01:/var/log\$' is visible at the bottom.

```
user root
Oct 14 11:10:54 TargetLinux01 sudo: user : TTY=pts/0 ; PWD=/ ; USER=root ;
COMMAND=/usr/sbin/fsck /dev/sdb5 -p
Oct 14 11:10:54 TargetLinux01 sudo: pam_unix(sudo:session): session opened for
user root by (uid=0)
Oct 14 11:10:55 TargetLinux01 sudo: pam_unix(sudo:session): session closed for
user root
Oct 14 11:11:05 TargetLinux01 sudo: user : TTY=pts/0 ; PWD=/ ; USER=root ;
COMMAND=/usr/sbin/fsck /dev/sdb5
Oct 14 11:11:05 TargetLinux01 sudo: pam_unix(sudo:session): session opened for
user root by (uid=0)
Oct 14 11:11:05 TargetLinux01 sudo: pam_unix(sudo:session): session closed for
user root
Oct 14 11:17:01 TargetLinux01 CRON[3007]: pam_unix(cron:session): session opene
d for user root by (uid=0)
Oct 14 11:17:01 TargetLinux01 CRON[3007]: pam_unix(cron:session): session close
d for user root
Oct 14 11:17:42 TargetLinux01 sudo: user : TTY=pts/0 ; PWD=/var/log ; USER=
root ; COMMAND=/usr/bin/tail -f kern.log
Oct 14 11:17:42 TargetLinux01 sudo: pam_unix(sudo:session): session opened for
user root by (uid=0)
Oct 14 11:18:31 TargetLinux01 sudo: pam_unix(sudo:session): session closed for
user root
Oct 14 11:18:50 TargetLinux01 sudo: user : TTY=pts/0 ; PWD=/var/log ; USER=
root ; COMMAND=/usr/bin/more -f auth.log
Oct 14 11:18:50 TargetLinux01 sudo: pam_unix(sudo:session): session opened for
user root by (uid=0)
user@TargetLinux01:/var/log$
user@TargetLinux01:/var/log$
```

Section 2: Applied Learning

Part 1: Identify Login Attempts on a Linux Drive Image

15. **Document** the names of the two non-root users that attempted to log in, the number of attempts detected, the date/time range of the attempts, the source IP address for the login attempts, and the port.

Name: noel

Number of attempts: 10 Date and Time: June 11, 00:57:11 -> 00:57:35, June 11, 05:06:38 -> 05:06:50

Source IP: 192.168.78.1 Port: 14444, 3521

Name: dominic Number of attempts: 8 Date and Time: June 11, 05:07:57 -> 05:07:57, June 11, 05:36:44 -> 05:39:01 Source IP: 192.168.78.1 Port: 4663, 3417

17. **Document** the date and time the most recent successful login for the user(s) that you previously identified in step 15.

User: Dominic Date and Time of most recent successful login: June 9, 13:31:59 and June 11, 05:23:03

User: noel No successful login identified.

Part 2: Identify Software Installations on a Linux Drive Image

3. **Document** the applications that were installed using apt-get, then use the Internet to identify the ones that might be considered suspicious.

Installed application: logkeys, autotools-dev, build-essential, autoconf, kbd, cacti Suspicious application installed might be logkeys (keylogger might have been used in spying password).

Part 3: Identify External Drive Attachments on a Linux Drive Image

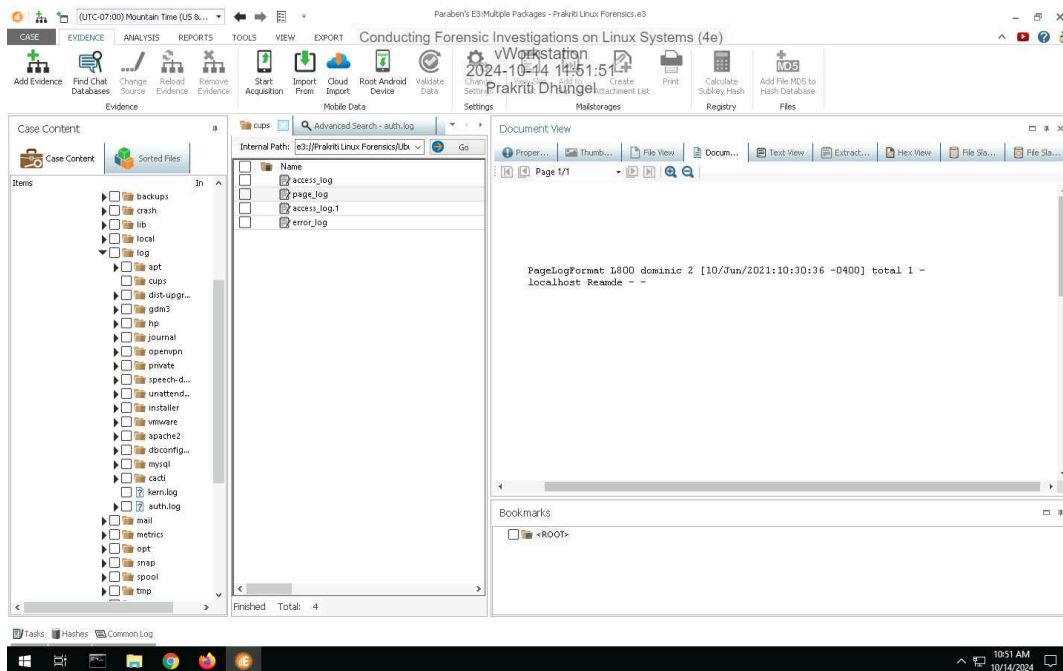
4. **Document** when the USB storage device was connected and its serial number.

USB Storage device connected - June 10, 10:24:12 Serial Number: FBI1405291710344

Section 3: Challenge and Analysis

Part 1: Identify Recently Printed Files on a Linux Drive Image

Make a screen capture showing the contents of the printer log file.



Part 2: Identify Disk Imaging on a Linux Drive Image

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Make a screen capture showing the record of the dd command in the Text View.

